

CM-001-M1-VS03A

Business Entity Registry & CRUD

Runtime Runbook

Objective

Implement the complete Business Entity Registry & CRUD feature for the Catrack Application Platform (CAP). Extend the existing metadata platform without redesigning completed architecture. Deliver production-ready implementation with no placeholders, TODOs, or mock logic.

Purpose

Provide a metadata-driven registry for business entity definitions such as Customer, Event, Employee, Vehicle, Menu, Recipe, Department and Supplier. This slice manages entity definitions only, not runtime records or field definitions.

Functional Scope

Entity List, Entity Details, Create, Edit, Archive, Restore, Delete (where permitted), Duplicate, Validation, Search, Filtering, Bulk Actions, Audit integration, Authorization integration. Publishing is excluded and belongs to VS03B.

Entity Metadata

Identity: Entity Name, Display Name, Plural Display Name, Entity Code, Description, Platform Module, Category. Runtime: Draft Status, Active, System Entity, Custom Entity, Metadata Locked. Navigation: Show in Navigation, Menu Group, Menu Order, Icon, Route. Behavior: Allow Create/Edit/Delete/Archive/Duplicate/Import/Export. Features: Searchable, Global Search, Auditable, Attachments, Notes, Activities, Tags, Workflow, Approval. API: API Enabled, API Name.

Entity List

AG Grid Enterprise with search, filtering, sorting, pagination, column chooser, bulk selection, export and refresh. Toolbar includes New, Edit, Duplicate, Archive, Restore, Delete, Export and View Audit.

Entity Details

Tabbed interface: General, Navigation, Runtime, API, Advanced and Audit.

Validation

Validate required fields, uniqueness of Entity Code, Entity Name, API Name and Route, reserved keywords, metadata lock, system entity protection and duplicate names within a module. Validation must execute on both client and server.

Authorization

Integrate with platform authorization. Permissions: Entity.View, Create, Edit, Delete, Archive, Restore, Duplicate and Export. Unauthorized actions must be hidden or blocked.

Audit

Audit Create, Update, Delete, Archive, Restore and Duplicate with user, timestamp, changed fields, previous values and new values.

Repository & Services

Implement repository methods for CRUD, archive, restore, duplicate and existence checks. Implement EntityService, EntityValidationService and EntityDuplicateService.

API & Server Actions

Implement route handlers and server actions for CRUD, archive, restore and duplicate using existing platform standards.

React Query

Implement useEntities, useEntity, useCreateEntity, useUpdateEntity, useArchiveEntity, useRestoreEntity, useDeleteEntity and useDuplicateEntity with proper cache invalidation.

Performance & Quality

Use server-side paging, filtering and sorting, efficient Prisma queries, transactions where required and avoid N+1 queries.

Testing

Repository, service, validation and API tests covering success and failure scenarios.

Completion Checklist

No TODOs, no placeholder code, no mock implementations, lint passes, build passes, prisma validate passes, repository/service/API/server actions integrated, authorization enforced and audit logging implemented.